

Marc-Aurèle RIVIÈRE

Data Science | Software Engineering | AI

📍 Trondheim, Norway

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7 years of data science and full-stack web development in consulting and research roles, with a focus on R. End-to-end delivery of production-grade analytics platforms across multiple industries, from design to deployment, with advanced statistical modeling and IaC-driven DB and cloud infrastructure management.

Experience

Data Scientist / R Developer

Remote

Data Champ'

2024 - 2025

Delivered data products end-to-end for clients across multiple industries, from requirements and architecture, through deployment and iteration, interfacing with both technical and non-technical stakeholders. Projects included:

- An investment analytics platform for France's development agency (AFD), to monitor their portfolio of billions in international development funding, with multi-database integration, interactive geospatial visualizations, and automated impact reporting.
- An enterprise-scale energy Measurement & Verification API for the Belgian Government, modeling energy consumption to predict the impact of renovations, with multi-platform data integration pipelines and automated report generation.
- A comprehensive HR management platform for the AFD to track thousands of employees across multiple branches.
- An integrated construction site management platform with cost and labor estimation, multi-format document parsing using LLMs, bidirectional ERP integration, automated quote generation, and smart project scheduling.
- A modular package for a branch of the French government, to integrate multiple sources of heterogeneous geo-spatial data, process them into spatio-temporal indicators at various levels of spatial granularity, and archive them in a specially-designed DB.

Data Science Consultant

Remote

Freelance

2023 - 2024

Sales/Marketing:

- Built end-to-end agentic pipelines for prospect discovery, enrichment, and personalized outreach, with custom ETL pipelines and MCP servers to integrate multiple CRMs, intent/signals providers, and social media platforms (R/Python/JS/LLM APIs)
- Designed and ran A/B tests comparing outreach strategies across prospect segments, to iterate on targeting and messaging

Bioinformatics:

- Data analysis for 4 published neuroscience papers: hierarchical GLMMs across heterogeneous data types (IHC, imaging, qPCR, behavioral); AR1 structures for time-varying repeated measures; multiple candidate models with PPC & AIC-based selection
- Created apps to allow researchers to visualize and analyze their data, with automatic enrichment from NCBI (R/Shiny)
- Designed automated pipelines to monitor new publications, protocols, and funded projects across multiple sources (LLM APIs)
- Built interactive dashboards to showcase results for select publications (R, Quarto, JS)

Research Engineer

Rouen, France

LITIS

2021 - 2023

- Designed & developed an Augmented Reality platform with motion tracking to evaluate and train non-visual navigation (Unity/C#)
- Designed a wearable vibrotactile belt and an interactive haptic tablet for haptic-based navigation for blind people (Java/Arduino)
- Designed psychophysics experiments to evaluate the impact of spatial cues on navigation and analyzed the resulting data (R)
- Developed and tested camera-based solutions for indoor localization (Python/C++)
- Formulated & wrote a 600k€ research grant proposal (ANR project "SAM-Guide", 2021)

Graduate Teaching Fellow

Rouen, France

University of Rouen

2017 - 2020

Gave university lectures & practicals on Web Development (JS), GUI design (Java), Image Processing/Computer Vision (Python)

Research Engineer Intern

San Francisco, USA

SKERI

2018

Worked on an indoor localization tool using Visual Inertial Odometry, particle filtering, and object detection (Python, Swift, OpenCV)

Research Engineer Intern

Grenoble, France

LPNC & GIPSA-Lab

2016

Worked on an image-to-sound conversion app for blind image exploration (C++), iteratively improved based on users' performance

Education

PhD - Biomedical Engineering

Rouen, France

Normandy University (unfinished - COVID)

2017 - 2020

- Design, develop and evaluate wearable haptic devices to assist blind people with navigation tasks in augmented reality (Java)
- Design, execute, and analyze multiple psychophysics experiments (R)
- Implement algorithms for real-time localization and mapping using embedded/mobile sensors (Python/C++)

MSc. - Computational Neurosciences

Grenoble, France

PHELMA - INPG

2015 - 2016

- Fundamentals of Memory, Perception, Linguistics, Imaging (e.g. fMRI, EEG, MEG), and Robotics
- Experimental design, Statistics, Bayesian modeling, Deep Learning, Signal processing (MATLAB & R)

Skills

Programming

R, SQL, Python, JS, Stan

Web Development

Shiny, plumber/2, Flask, HTMX, Bootstrap, Playwright

R Development

Shiny, bslib, shinytest2, tidyverse, data.table, renv, ggplot2, plotly/ggiraph, DBI/pool, brms, stan-verse, glmmTMB, easystats, DHARMa, plumber2, roxygen2/pkgdown, Quarto

Statistics

Experimental Design, Inference, A/B testing, GLMM, GAM, GEE, Bayesian Modeling

DBMS

Postgres, DuckDB, SQLite

DevOps / Infra

Git, Unix, Docker, CI/CD, Ansible, Terraform

LLM / Agentic

LLM APIs, MCP servers, Custom agents, RAG, Local inference, AI-assisted development

Languages

French (Native), English (Bilingual), Norwegian (A2/B1), Spanish (A2)

Packages

Argent

 ma-riviere/Argent

R Package

A unified interface to build AI Agents using Large Language Models (LLMs) from multiple providers, supporting tool calling (functions, server tools, and MCP servers), multimodal inputs, and structured outputs.

Publications

- Rodriguez-Duboc, A., Basille-Dugay, M., Debonne, A., **Rivière, M.-A.**, Vaudry, D., & Burel, D. (2023). Apnea of prematurity induces short and long-term development-related transcriptional changes in the murine cerebellum. *Current Research in Neurobiology*, 5, 100113.
- Faugloire, E., Lejeune, L., **Rivière, M.-A.**, & Mantel, B. (2022). Spatiotemporal influences on the recognition of two-dimensional vibrotactile patterns on the abdomen. *Journal of Experimental Psychology: Applied*, 28(3), 606–628.
- Coughlan, J.M., Biggs, B., **Rivière, M.-A.**, Shen, H. (2020). An Audio-Based 3D Spatial Guidance AR System for Blind Users. In *Lecture Notes in Computer Science* (Vol. 12376, pp. 475–484). Springer.
- **Rivière, M.-A.**, Gay, S., Romeo, K., Pissaloux, E., Bujacz, M., Skulimowski, P., & Strumillo, P. (2019). NAV-VIR: An audio-tactile virtual environment to assist visually impaired people. *Proceedings of the 9th International IEEE/EMBS Conference on Neural Engineering*, 1038–1041.
- **Rivière, M.-A.**, Gay, S., Pissaloux, E. (2018). TactiBelt: Integrating Spatial Cognition and Mobility Theories into the Design of a Novel Orientation and Mobility Assistive Device for the Blind. In *Lecture Notes in Computer Science* (Vol. 10897, pp. 110–113). Springer.
- Gay, S., **Rivière, M.-A.**, Pissaloux, E. (2018). Towards Haptic Surface Devices with Force Feedback for Visually Impaired People. In *Lecture Notes in Computer Science* (Vol. 10897, pp. 258–266). Springer.

Data analysis lead in:

- Rodriguez-Duboc, A., Racine, C., Basille-Dugay, M., Vaudry, D., Gonzalez, B., & Burel, D. (2026). Innovative 3D-Image Analysis of Cerebellar Vascularization Highlights Angiogenic Gene Dysregulations in a Murine Model of Apnea of Prematurity. *The Cerebellum*, 25(3), 62.
- Rodriguez-Duboc, A., Velicer, S. T., & Kibro-Flatmoen, A. (2025). Bnip3 expression in the brain of an Alzheimer's disease rat model. *Brain Mechanisms*, 148–150, 202518.
- Leroux, S., Rodriguez-Duboc, A., Arabo, A., Basille-Dugay, M., Vaudry, D., & Burel, D. (2022). Intermittent hypoxia in a mouse model of apnea of prematurity leads to a retardation of cerebellar development and long-term functional deficits. *Cell & Bioscience*, 12(1), 148.